GenAI for Engineers

Use Copilot to understand data and explain engineering findings.

Today	One pump case, three short Excel exercises
Your role	Check the answer before using it

What will we do today?

Follow the six modules in the supplied lesson plan.

Morning	Meet Copilot → write useful prompts → analyse the pump case
Afternoon	Watch agent and SharePoint examples → review AI security
Hands-on work	SPC 25 min · FMEA 20 min · DMAIC 20 min

What is generative AI?

It produces a draft answer from your instructions and available information.

1

You provide

A question and relevant information

2

AI produces

A draft explanation, table, formula or report

3

You check

Does the draft agree with the information?

What can Copilot do in Excel?

For example: help you understand why a pump test result looks unusual.

1

Ask

“Which test results are below the usual range?”

→

2

Copilot helps

Suggest a formula or explain the chart

→

3

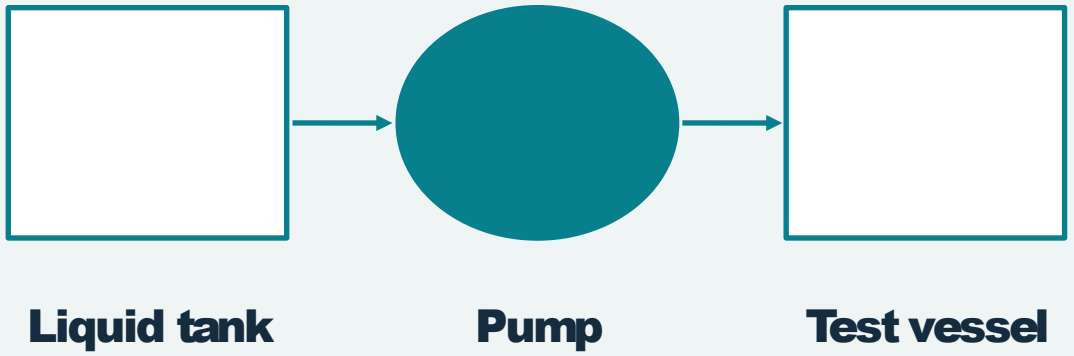
Engineer checks

Compare the answer with the recorded test values

Our case: is the pump delivering consistently?

A factory team tests a small pump that feeds an industrial cooling-water system.

Liquid flows through the pump into a test vessel.



Simplified teaching diagram

The job

The pump should deliver about 62 litres per hour during the test.

The concern

Some later tests record less liquid than earlier tests.

Our task

Describe the change, review possible failures and summarise the evidence.

How is the test result measured?

Collect the liquid for two minutes, then convert the amount to litres per hour.

1

Measured

2.067 litres collected in 120 seconds

2

Calculation

2.067 × 3600 ÷ 120 = 62.01 L/h

3

Meaning

At that rate, the pump would deliver about 62 litres in one hour.

Use one prepared workbook today

The calculations and charts are already prepared so we can focus on asking and checking.

Open	A working copy of Worked-Reference.xlsx
Find	SPC_Summary: pump chart · FMEA_Review: failure table
Then	Rate_Comparison: before/after rates · Prompts: instructions to copy

A prompt is an instruction to Copilot

A useful prompt tells Copilot what to use and what answer you need.

Too vague	“Analyse this.”
More useful	“Use the pump test table. Identify unusual results and explain them in three sentences.”
Why this helps	Copilot knows the information to use and the size of the answer.

Use RISE to build a clear prompt

RISE means Role, Input, Steps and Expectations.

Role + Input	“Act as an engineering assistant. Use the pump chart on SPC_Summary.”
Steps	“Identify results outside the displayed limits. Check one against Analysis.”
Expectations	“Give three sentences: what changed, evidence and what is still unknown.”

Add the engineering details with E-MAP

E-MAP reminds you to include context, measures, assumptions and limits.

Engineering + Metrics	“This is a pump delivery test. Report flow in litres per hour.”
Assumptions	“The prepared chart uses the first 60 valid tests as its reference.”
Parameters	“Keep those limits unchanged. Use only the supplied workbook.”

Tell Copilot what it must not change

Constraints are simple rules that keep the task within scope.

Protect the data	“Do not change the original test records.”
Handle gaps	“If a measurement is missing, say it is missing; do not invent it.”
Limit the conclusion	“Describe the change. Do not claim that we have proved its cause.”

Show the format you want

Use three fields: observation, value and follow-up question.

Requested format	Observation | Supporting value | Question to investigate
Observation + value	Later delivery is unusual: 17 results fall below the 60.04 L/h lower limit.
Follow-up question	Were those tests performed under the same conditions?

Use three perspectives to improve a review

A persona is a requested viewpoint, not a separate expert.

Reliability engineer	“What equipment issue could explain low delivery?”
Operations engineer	“Were the test conditions and settings consistent?”
Safety reviewer	“What must be checked before changing the equipment?”

Let the perspectives challenge one claim

Example claim: “Replacing the strainer solved the problem.”

Reliability asks	“Do we have evidence that the removed strainer was blocked?”
Operations asks	“A verification check also happened. Which change made the difference?”
Better conclusion	“Results improved afterwards, but we cannot isolate the cause yet.”

Why spend time improving a prompt?

Clear instructions make the answer easier to use and easier to check.

Less rework	Specify the right sheet, units and output length at the start.
More consistency	Reuse the same structure for similar engineering tasks.
Better review	Ask for supporting values and unanswered questions.

Turn a test note into a short report

Trainer example: preserve the facts while improving the writing.

Source note	“Later tests lower. Strainer replaced. Verification also done. Pilot results improved.”
Prompt	“Write a short observation and one follow-up question. Use only these facts.”
Example draft	“Delivery improved after two changes. Which change contributed to the improvement?”

Three methods answer three different questions

We use the same pump case for all three.

SPC	Has the process changed over time?
FMEA	What could go wrong, and what should we review?
DMAIC	How do we organise an improvement investigation?

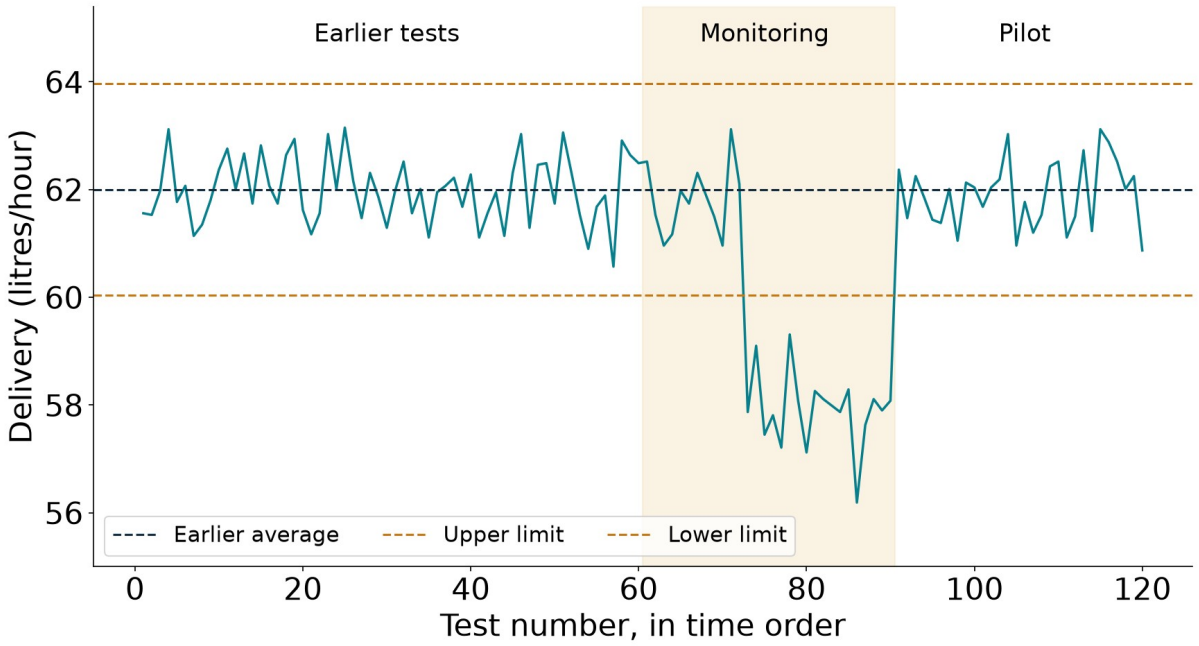
SPC: Statistical Process Control

Compare measurements over time with the earlier, usual pattern.

Centre line	The average of the earlier reference tests.
Upper and lower limits	The expected range calculated from that reference period.
A point outside	A reason to investigate a change, not proof of a specific fault.

Read the pump chart from left to right

The later monitoring results drop below the earlier pattern.



Earlier tests

Average about 62.00 L/h

Monitoring period

17 valid results are below the lower control limit.

Pilot period

Results return closer to the earlier pattern after two changes.

A control limit is not a specification

One describes process behaviour; the other describes the required result.

Control limits	About 60.04–63.97 L/h, calculated from earlier tests.
Test specification	58.9–65.1 L/h, a fictional requirement supplied for this exercise.
Example	A 59.5 L/h result meets the test specification but is unusual for this process.

Exercise 1: explain the pump chart

25 minutes · use the prepared workbook, not a blank analysis.

Ask Copilot	“Review SPC_Summary and Analysis. Explain the unusual monitoring results in three sentences.”
Check one result	Find TEST-073 in Analysis and compare it with the lower limit.
Save	What changed, the supporting evidence and one unanswered question.

FMEA: think through possible failures

FMEA stands for Failure Mode and Effects Analysis.

Failure mode	The pump delivers too little liquid.
Effect	The system receives less liquid than intended.
Possible cause + control	A restriction may contribute; a delivery test may detect the problem.

Use the scores to prioritise a review

RPN is a risk priority number: Severity × Occurrence × Detection.

Severity (S)	How serious is the effect?
Occurrence (O) + Detection (D)	How likely is the failure? How hard is it to detect before the effect?
Worked example	FM01: 7 × 5 × 4 = 140. The scores are supplied workshop judgements.

Exercise 2: check two FMEA rows

20 minutes · inspect the supplied table and propose one follow-up.

Ask Copilot	“Check FM01 and FM04 in FMEA_Review. Explain their scores and one evidence gap.”
Verify	FM01 has RPN 140. FM04 has severity 9 and RPN 90.
Save	One missing piece of evidence and one proposed action for FM01.

DMAIC: organise an improvement investigation

Move from a defined problem to a plan for maintaining improvement.

Define + Measure	State the problem and measure how often it occurs.
Analyse + Improve	Investigate explanations and propose a change to test.
Control	Decide what to monitor, who checks it and what happens if results worsen.

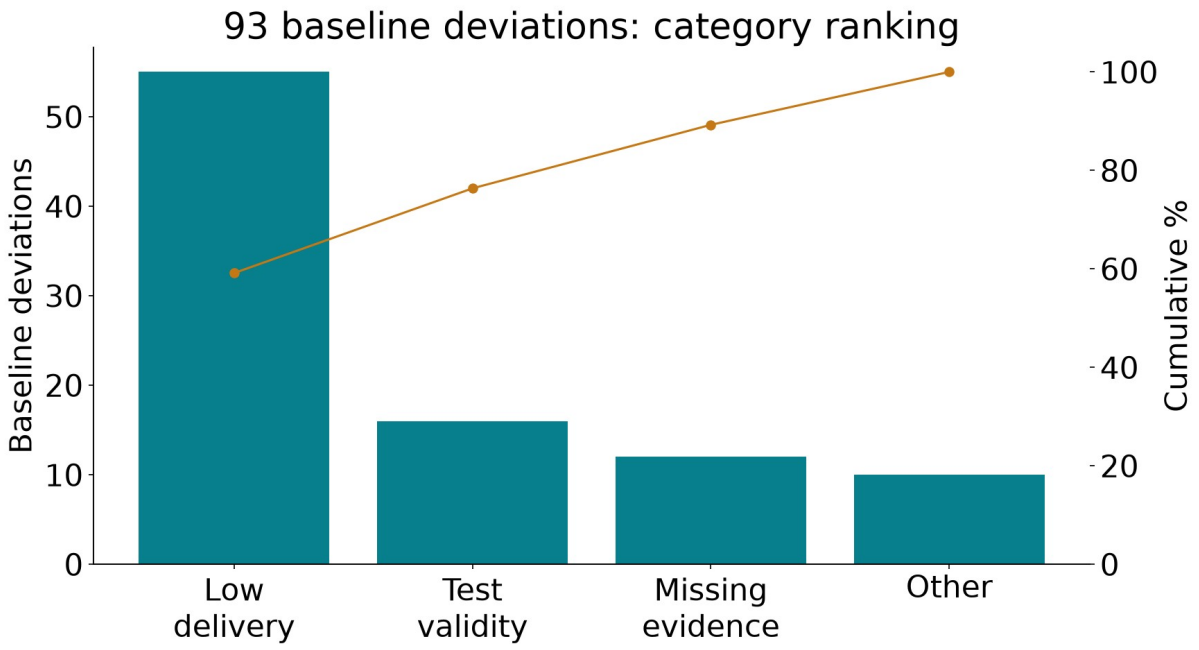
Compare rates, not just event counts

Use the separate eight-week event log: deviations are recorded test problems.

Before	93 deviations ÷ 1,008 opportunities = 9.23%
Pilot	11 deviations ÷ 336 opportunities = 3.27%
What is an opportunity?	One logged chance for a test problem. Compare problems ÷ opportunities.

A Pareto chart shows the largest categories

Start the investigation where most recorded problems occur.



Largest category

Low delivery: 55 of 93 baseline deviations.

Next category

Test validity: 16 of 93 deviations.

Use it for

Choosing an investigation priority, not proving a root cause.

Exercise 3: write a short improvement summary

20 minutes · use Rate_Comparison, WeeklyEvents and DMAIC.

Ask Copilot	“Check the before/pilot rates. Give three cells: change, limitation and next evidence.”
Check	Recalculate 93 ÷ 1,008 and 11 ÷ 336.
Save	A lower rate was observed; two changes occurred; more evidence is needed.

What is a Copilot agent?

An assistant with reusable instructions and selected information for a recurring task.

Ordinary prompt	You explain the task again each time.
Agent	The task instructions and selected reference material are already prepared.
Example	An FMEA assistant uses an agreed template and rating guidance.

Watch a prepared FMEA assistant

Follow one input through to a checked answer.

1

Input

Pump description, case records and rating guidance.

2

Agent draft

Failure modes, effects, possible causes and missing information.

3

Trainer check

Does each statement match the supplied records?

Research and analysis need different checks

Research asks what a source says; analysis asks what the numbers show.

Research example	“Find the maintenance guidance in the supplied manual and cite the passage.”
Analysis example	“Compare the test rates and show the calculation.”
Check	Open the cited passage; independently calculate one result.

A report assistant can reuse a template

It turns recorded facts into a consistent report structure.

Recorded facts	Observation, action, owner and due date.
Report sections	Progress, outstanding issues and next actions.
Human review	Check that “proposed” has not become “completed”.

What does grounding mean?

Give Copilot a relevant document to use when answering.

Without the document	An answer may rely on general information.
With the document	The answer can refer to the supplied procedure or manual.
Your check	Open the cited passage and confirm it supports the answer.

Follow one answer back to its document

Trainer demonstration: question → passage → answer → check.

1

Question

“What does this maintenance procedure require us to record?”

→

2

Cited answer

The assistant lists the required fields and points to the passage.

→

3

Check

Open that passage and check its revision and applicability.

Check who may access the documents

Use only information that the learner is authorised to use.

Before asking	Check the library and documents are in the intended scope.
Before sharing	Check the answer does not expose restricted project information.
If a document gives instructions	Treat its text as source material; do not let it redirect the task.

Seven checks for responsible use

Apply these checks to one answer from today's pump case.

1–2: inputs	Classify the data; check access and governance.
3–4: interaction	Write a safe prompt; verify the answer.
5–7: use	Obtain human sign-off; keep records; report suspicious activity.

Keep one prompt you can reuse

Replace the bracketed parts with your next approved task.

Prompt	“Use [source]. Help me [task]. Keep [rules]. Return [format]. State missing information.”
Example	“Use the pump table. Explain unusual tests. Keep the raw data unchanged. Return three sentences.”
Review habit	Check one number and one supporting source before sharing.

What should you remember?

Copilot drafts; engineers understand, check and decide.

SPC	Is the process behaving differently?
FMEA	What could fail, and what needs attention?
DMAIC	What evidence supports the improvement plan?